

CareerIQ Database — Detailed 25-Table Schema

MySQL Workbench implementation specification for the current MCA prototype

Purpose of this document

This document converts the uploaded CareerIQ Database Design Guide into a practical table-by-table schema that can be created in MySQL Workbench. It lists the recommended columns, MySQL data types, primary keys, foreign keys, unique constraints, nullability, defaults, and the reason each column is present.

Source boundary: The Database Design Guide explicitly defines the 25 tables, their relationships, and the information each table stores, but it does not provide a complete column-by-column SQL schema. Therefore, columns and implementation details marked here as recommended are database-design decisions needed to turn the guide into an executable MySQL schema. They should be reviewed by the project team before final deployment.

Current prototype: The design is for the controlled dummy-data version. Live job portal/API tables and future what-if/market tables are intentionally excluded.

Global MySQL rules

Rule	Implementation
Storage engine	InnoDB
Character set	utf8mb4
Foreign keys	Create all relationships listed in this document
Scores / percentages	DECIMAL(5,2) unless a larger contribution/priority range is needed
Skill levels	TINYINT UNSIGNED, intended range 1–5
Passwords	Store password_hash only; never store plain passwords
Resumes	Store file path/URL and extracted text; do not store the resume file as BLOB
Master data	Avoid deleting skills/career roles that are referenced; use active_status

25-table architecture at a glance

- 1. Candidate Profile** — users → candidate_profiles → preferences/education/experience/projects/certifications/resumes
- 2. Skills** — skills → skill_aliases; candidate_profiles → candidate_skills → skills
- 3. Reference Dataset** — career_roles → career_role_skills → skills; jobs → job_skills → skills
- 4. Analysis Results** — analysis_runs → career recommendations, job matches, explanations, readiness, skill gaps
- 5. Learning** — learning_resources ↔ skills; analysis → learning_roadmaps → roadmap_steps

1. users

Module: 1. Candidate Profile

Purpose: Account and login data.

Indexes / constraints: UNIQUE(email)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
user_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Unique account identifier
full_name	VARCHAR(100)	—	NO	—	—	User's full name
email	VARCHAR(255)	UQ	NO	—	—	Login email; must be unique
password_hash	VARCHAR(255)	—	NO	—	—	Store a password hash only, never plain password
account_role	VARCHAR(30)	—	NO	—	'candidate'	Account role such as candidate/admin
active_status	TINYINT(1)	—	NO	—	1	1 = active, 0 = inactive
created_at	DATETIME	—	NO	—	CURRENT_TIMESTAMP	Account creation time
updated_at	DATETIME	—	NO	—	CURRENT_TIMESTAMP	Last account update time

2. candidate_profiles

Module: 1. Candidate Profile

Purpose: Main candidate profile details. One profile belongs to one user.

Indexes / constraints: UNIQUE(user_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
profile_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Profile identifier
user_id	BIGINT UNSIGNED	FK + UQ	NO	users.user_id	—	One-to-one link to users
phone	VARCHAR(20)	—	YES	—	NULL	Candidate phone number
city	VARCHAR(100)	—	YES	—	NULL	Candidate city
linkedin_url	VARCHAR(500)	—	YES	—	NULL	LinkedIn profile URL
portfolio_url	VARCHAR(500)	—	YES	—	NULL	Portfolio URL
profile_summary	TEXT	—	YES	—	NULL	Candidate summary
created_at	DATETIME	—	NO	—	CURRENT_TIMESTAMP	Profile creation time
updated_at	DATETIME	—	NO	—	CURRENT_TIMESTAMP	Last profile update

Foreign-key relationships

user_id → users.user_id

3. candidate_preferences

Module: 1. Candidate Profile

Purpose: Candidate's target role and job preferences.

Indexes / constraints: UNIQUE(profile_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
preference_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Preference identifier
profile_id	BIGINT UNSIGNED	FK + UQ	NO	candidate_profiles.profile_id	—	Profile owning these preferences
target_role_id	BIGINT UNSIGNED	FK	YES	career_roles.career_role_id	NULL	Preferred career role
preferred_location	VARCHAR(150)	—	YES	—	NULL	Preferred work location
work_mode	VARCHAR(30)	—	YES	—	NULL	Remote, onsite, hybrid, etc.
expected_salary	DECIMAL(12,2)	—	YES	—	NULL	Expected salary amount
salary_currency	CHAR(3)	—	YES	—	'INR'	Currency code
created_at	DATETIME	—	NO	—	CURRENT_TIME STAMP	Preference creation time
updated_at	DATETIME	—	NO	—	CURRENT_TIME STAMP	Last preference update

Foreign-key relationships

profile_id → candidate_profiles.profile_id

target_role_id → career_roles.career_role_id

4. education

Module: 1. Candidate Profile

Purpose: Academic records.

Indexes / constraints: INDEX(profile_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
education_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Education record identifier
profile_id	BIGINT UNSIGNED	FK	NO	candidate_profiles.profile_id	—	Candidate profile
degree	VARCHAR(150)	—	NO	—	—	Degree name
college	VARCHAR(200)	—	NO	—	—	College/institution
specialization	VARCHAR(150)	—	YES	—	NULL	Specialization/major
score	DECIMAL(5,2)	—	YES	—	NULL	CGPA or percentage value
score_type	VARCHAR(20)	—	YES	—	NULL	CGPA or percentage
start_year	SMALLINT UNSIGNED	—	YES	—	NULL	Academic start year
end_year	SMALLINT UNSIGNED	—	YES	—	NULL	Academic end year

Foreign-key relationships

profile_id → candidate_profiles.profile_id

5. experience

Module: 1. Candidate Profile

Purpose: Internship or work records.

Indexes / constraints: INDEX(profile_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
experience_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Experience record identifier
profile_id	BIGINT UNSIGNED	FK	NO	candidate_profiles.profile_id	—	Candidate profile
company	VARCHAR(200)	—	NO	—	—	Company/organization
designation	VARCHAR(150)	—	NO	—	—	Job title/designation
start_date	DATE	—	NO	—	—	Experience start date
end_date	DATE	—	YES	—	NULL	End date; NULL if current
is_current	TINYINT(1)	—	NO	—	0	1 = current experience
description	TEXT	—	YES	—	NULL	Experience description

Foreign-key relationships

profile_id → candidate_profiles.profile_id

6. projects

Module: 1. Candidate Profile

Purpose: Candidate project records.

Indexes / constraints: INDEX(profile_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
project_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Project identifier
profile_id	BIGINT UNSIGNED	FK	NO	candidate_profiles.profile_id	—	Candidate profile
project_title	VARCHAR(200)	—	NO	—	—	Project title
description	TEXT	—	YES	—	NULL	Project description
github_url	VARCHAR(500)	—	YES	—	NULL	GitHub URL
live_url	VARCHAR(500)	—	YES	—	NULL	Live project URL
technologies	TEXT	—	YES	—	NULL	Technologies used
start_date	DATE	—	YES	—	NULL	Project start date
end_date	DATE	—	YES	—	NULL	Project end date

Foreign-key relationships

profile_id → candidate_profiles.profile_id

7. certifications

Module: 1. Candidate Profile

Purpose: Courses and certificate records.

Indexes / constraints: INDEX(profile_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
certification_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Certification identifier
profile_id	BIGINT UNSIGNED	FK	NO	candidate_profiles.profile_id	—	Candidate profile
certificate_name	VARCHAR(200)	—	NO	—	—	Certificate/course name
issuer	VARCHAR(200)	—	NO	—	—	Issuing organization
issue_date	DATE	—	YES	—	NULL	Certificate issue date
credential_url	VARCHAR(500)	—	YES	—	NULL	Credential verification URL

Foreign-key relationships

profile_id → candidate_profiles.profile_id

8. resumes

Module: 1. Candidate Profile

Purpose: Uploaded resume metadata and extracted text.

Indexes / constraints: INDEX(profile_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
resume_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Resume identifier
profile_id	BIGINT UNSIGNED	FK	NO	candidate_profiles.profile_id	—	Candidate profile
file_name	VARCHAR(255)	—	NO	—	—	Original/stored file name
storage_path	VARCHAR(1000)	—	NO	—	—	File path or URL; do not store PDF as BLOB
extracted_text	LONGTEXT	—	YES	—	NULL	Text extracted from resume
parser_status	VARCHAR(30)	—	NO	—	'pending'	pending/success/failed
uploaded_at	DATETIME	—	NO	—	CURRENT_TIMESTAMP	Upload timestamp

Foreign-key relationships

profile_id → candidate_profiles.profile_id

9. skills

Module: 2. Skills

Purpose: Master list of normalized skills.

Indexes / constraints: UNIQUE(normalized_name)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
skill_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Skill identifier
skill_name	VARCHAR(150)	—	NO	—	—	Display name, e.g. Power BI
normalized_name	VARCHAR(150)	UQ	NO	—	—	Normalized matching name
description	TEXT	—	YES	—	NULL	Optional skill description
active_status	TINYINT(1)	—	NO	—	1	1 = active master skill
created_at	DATETIME	—	NO	—	CURRENT_TIMESTAMP	Creation time

10. skill_aliases

Module: 2. Skills

Purpose: Alternative names used to identify the same skill.

Indexes / constraints: UNIQUE(alias_name); INDEX(skill_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
alias_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Alias identifier
skill_id	BIGINT UNSIGNED	FK	NO	skills.skill_id	—	Master skill
alias_name	VARCHAR(150)	UQ	NO	—	—	Alternative name such as PowerBI or Python3
normalized_alias	VARCHAR(150)	—	NO	—	—	Normalized alias for matching

Foreign-key relationships

skill_id → skills.skill_id

11. candidate_skills

Module: 2. Skills

Purpose: Candidate skill ownership and proficiency.

Indexes / constraints: UNIQUE(profile_id, skill_id); INDEX(skill_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
candidate_skill_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Candidate-skill record identifier
profile_id	BIGINT UNSIGNED	FK	NO	candidate_profiles.profile_id	—	Candidate profile
skill_id	BIGINT UNSIGNED	FK	NO	skills.skill_id	—	Master skill
proficiency_level	TINYINT UNSIGNED	—	NO	—	1	Candidate level from 1 to 5
experience_months	SMALLINT UNSIGNED	—	YES	—	0	Months of experience
source	VARCHAR(20)	—	NO	—	'manual'	manual or resume
extraction_confidence	DECIMAL(5,2)	—	YES	—	NULL	Confidence percentage when extracted
created_at	DATETIME	—	NO	—	CURRENT_TIME STAMP	Record creation time

Foreign-key relationships

profile_id → candidate_profiles.profile_id

skill_id → skills.skill_id

12. career_roles

Module: 3. Reference Dataset

Purpose: Available career paths.

Indexes / constraints: UNIQUE(role_name)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
career_role_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Career role identifier
role_name	VARCHAR(150)	UQ	NO	—	—	Role name such as Data Analyst
description	TEXT	—	YES	—	NULL	Career role description
active_status	TINYINT(1)	—	NO	—	1	1 = active role
created_at	DATETIME	—	NO	—	CURRENT_TIMESTAMP	Creation time

13. career_role_skills

Module: 3. Reference Dataset

Purpose: Skills required for each career role.

Indexes / constraints: UNIQUE(career_role_id, skill_id); INDEX(skill_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
role_skill_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Role-skill record identifier
career_role_id	BIGINT UNSIGNED	FK	NO	career_roles.career_role_id	—	Career role
skill_id	BIGINT UNSIGNED	FK	NO	skills.skill_id	—	Required skill
required_level	TINYINT UNSIGNED	NO	NO	—	1	Required level from 1 to 5
importance_weight	DECIMAL(5,2)	—	NO	—	1.00	Weight used by matching logic
mandatory_flag	TINYINT(1)	—	NO	—	0	1 = mandatory skill

Foreign-key relationships

career_role_id → career_roles.career_role_id

skill_id → skills.skill_id

14. jobs

Module: 3. Reference Dataset

Purpose: Curated dummy job opportunities.

Indexes / constraints: INDEX(career_role_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
job_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Job identifier
career_role_id	BIGINT UNSIGNED	FK	NO	career_roles.career_role_id	—	Related career role
job_title	VARCHAR(200)	—	NO	—	—	Dummy job title
company_name	VARCHAR(200)	—	NO	—	—	Sample company
location	VARCHAR(150)	—	YES	—	NULL	Job location
work_mode	VARCHAR(30)	—	YES	—	NULL	Remote/onsite/hybrid
description	TEXT	—	YES	—	NULL	Job description
experience_requirement	VARCHAR(100)	—	YES	—	NULL	Experience requirement
is_demo_data	TINYINT(1)	—	NO	—	1	Prototype dummy-data flag
source	VARCHAR(100)	—	YES	—	NULL	Dataset/source label
created_at	DATETIME	—	NO	—	CURRENT_TIMESTAMP	Record creation time

Foreign-key relationships

career_role_id → career_roles.career_role_id

15. job_skills

Module: 3. Reference Dataset

Purpose: Skills required by each dummy job.

Indexes / constraints: UNIQUE(job_id, skill_id); INDEX(skill_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
job_skill_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Job-skill record identifier
job_id	BIGINT UNSIGNED	FK	NO	jobs.job_id	—	Job
skill_id	BIGINT UNSIGNED	FK	NO	skills.skill_id	—	Required skill
required_level	TINYINT UNSIGNED	—	NO	—	1	Required level from 1 to 5
importance_weight	DECIMAL(5,2)	—	NO	—	1.00	Weight used by matching logic
mandatory_flag	TINYINT(1)	—	NO	—	0	1 = mandatory skill

Foreign-key relationships

job_id → jobs.job_id

skill_id → skills.skill_id

16. analysis_runs

Module: 4. Analysis Results

Purpose: One saved profile analysis each time Analyze My Career is run.

Indexes / constraints: INDEX(profile_id); INDEX(resume_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
analysis_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Analysis identifier
profile_id	BIGINT UNSIGNED	FK	NO	candidate_profiles.profile_id	—	Profile being analyzed
resume_id	BIGINT UNSIGNED	FK	YES	resumes.resume_id	NULL	Resume used for analysis, if applicable
analysis_date	DATETIME	NO	NO	—	CURRENT_TIME STAMP	Analysis date/time
algorithm_version	VARCHAR(50)	—	NO	—	'v1.0'	Version of matching/AI logic
status	VARCHAR(30)	—	NO	—	'started'	started/completed/failed
started_at	DATETIME	—	NO	—	CURRENT_TIME STAMP	Analysis start
completed_at	DATETIME	—	YES	—	NULL	Analysis completion

Foreign-key relationships

profile_id → candidate_profiles.profile_id

resume_id → resumes.resume_id

17. career_recommendations

Module: 4. Analysis Results

Purpose: Career-role match results produced by an analysis.

Indexes / constraints: UNIQUE(analysis_id, career_role_id); INDEX(career_role_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
career_recommendation_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Recommendation identifier
analysis_id	BIGINT UNSIGNED	FK	NO	analysis_runs.analysis_id	—	Analysis run
career_role_id	BIGINT UNSIGNED	FK	NO	career_roles.career_role_id	—	Recommended role
match_percentage	DECIMAL(5,2)	—	NO	—	0.00	Career match percentage
recommendation_rank	INT UNSIGNED	—	NO	—	1	Rank within this analysis
summary_explanation	TEXT	—	YES	—	NULL	Short explanation

Foreign-key relationships

analysis_id → analysis_runs.analysis_id

career_role_id → career_roles.career_role_id

18. job_matches

Module: 4. Analysis Results

Purpose: Dummy-job match results produced by an analysis.

Indexes / constraints: UNIQUE(analysis_id, job_id); INDEX(job_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
job_match_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Job-match identifier
analysis_id	BIGINT UNSIGNED	FK	NO	analysis_runs.analysis_id	—	Analysis run
job_id	BIGINT UNSIGNED	FK	NO	jobs.job_id	—	Matched dummy job
match_percentage	DECIMAL(5,2)	—	NO	—	0.00	Job match percentage
match_rank	INT UNSIGNED	—	NO	—	1	Rank within this analysis
summary_explanation	TEXT	—	YES	—	NULL	Short explanation

Foreign-key relationships

analysis_id → analysis_runs.analysis_id

job_id → jobs.job_id

19. match_explanations

Module: 4. Analysis Results

Purpose: Detailed factors explaining a career or job match score.

Indexes / constraints: INDEX(analysis_id); INDEX(job_match_id); INDEX(career_recommendation_id); INDEX(skill_id); CHECK: exactly one of job_match_id or career_recommendation_id should be filled

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
explanation_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Explanation identifier
analysis_id	BIGINT UNSIGNED	FK	NO	analysis_runs.analysis_id	—	Analysis run
job_match_id	BIGINT UNSIGNED	FK	YES	job_matches.job_match_id	NULL	Use for job explanation
career_recommendation_id	BIGINT UNSIGNED	FK	YES	career_recommendations.career_recommendation_id	NULL	Use for career explanation
skill_id	BIGINT UNSIGNED	FK	YES	skills.skill_id	NULL	Skill involved in explanation
factor_type	VARCHAR(30)	—	NO	—	matched	matched_skill/missing_skill/etc
contribution	DECIMAL(7,2)	—	YES	—	NULL	Contribution to score
explanation_text	TEXT	—	NO	—	—	Human-readable reason

Foreign-key relationships

analysis_id → analysis_runs.analysis_id

job_match_id → job_matches.job_match_id

career_recommendation_id → career_recommendations.career_recommendation_id

skill_id → skills.skill_id

20. readiness_scores

Module: 4. Analysis Results

Purpose: Overall career preparation score and components.

Indexes / constraints: UNIQUE(analysis_id, career_role_id); INDEX(career_role_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
readiness_score_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Readiness record identifier
analysis_id	BIGINT UNSIGNED	FK	NO	analysis_runs.analysis_id	—	Analysis run
career_role_id	BIGINT UNSIGNED	FK	NO	career_roles.career_role_id	—	Target career role
overall_score	DECIMAL(5,2)	—	NO	—	0.00	Overall readiness percentage/score
skills_score	DECIMAL(5,2)	—	YES	—	NULL	Skills component
education_score	DECIMAL(5,2)	—	YES	—	NULL	Education component
experience_score	DECIMAL(5,2)	—	YES	—	NULL	Experience component
projects_score	DECIMAL(5,2)	—	YES	—	NULL	Projects component

Foreign-key relationships

analysis_id → analysis_runs.analysis_id

career_role_id → career_roles.career_role_id

21. skill_gaps

Module: 4. Analysis Results

Purpose: Missing or weak skills for a target career role.

Indexes / constraints: UNIQUE(analysis_id, career_role_id, skill_id); INDEX(skill_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
skill_gap_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Skill-gap identifier
analysis_id	BIGINT UNSIGNED	FK	NO	analysis_runs.analysis_id	—	Analysis run
career_role_id	BIGINT UNSIGNED	FK	NO	career_roles.career_role_id	—	Target role
skill_id	BIGINT UNSIGNED	FK	NO	skills.skill_id	—	Missing/weak skill
current_level	TINYINT UNSIGNED	—	NO	—	0	Candidate level, 0 if absent
required_level	TINYINT UNSIGNED	—	NO	—	1	Required role level
priority_score	DECIMAL(7,2)	—	NO	—	0.00	Priority used to order gaps
status	VARCHAR(20)	—	NO	—	'missing'	missing/weak/covered

Foreign-key relationships

analysis_id → analysis_runs.analysis_id

career_role_id → career_roles.career_role_id

skill_id → skills.skill_id

22. learning_resources

Module: 5. Learning

Purpose: Courses, videos, articles and other learning resources.

Indexes / constraints: INDEX(resource_type); INDEX(is_free)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
resource_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Resource identifier
title	VARCHAR(255)	—	NO	—	—	Course/video/article title
provider	VARCHAR(150)	—	YES	—	NULL	Resource provider
resource_type	VARCHAR(30)	—	NO	—	—	course/video/article/etc.
url	VARCHAR(1000)	—	NO	—	—	Resource URL
duration_minutes	INT UNSIGNED	—	YES	—	NULL	Duration in minutes
is_free	TINYINT(1)	—	NO	—	1	1 = free, 0 = paid
active_status	TINYINT(1)	—	NO	—	1	1 = active resource
created_at	DATETIME	—	NO	—	CURRENT_TIMESTAMP	Creation time

23. learning_resource_skills

Module: 5. Learning

Purpose: Bridge table showing which skills each learning resource teaches.

Indexes / constraints: PRIMARY KEY(resource_id, skill_id); INDEX(skill_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
resource_id	BIGINT UNSIGNED	PK + FK	NO	learning_resources.resource_id	—	Part of composite primary key
skill_id	BIGINT UNSIGNED	PK + FK	NO	skills.skill_id	—	Part of composite primary key

Foreign-key relationships

resource_id → learning_resources.resource_id

skill_id → skills.skill_id

24. learning_roadmaps

Module: 5. Learning

Purpose: Personalized roadmap header for an analysis and target career role.

Indexes / constraints: INDEX(analysis_id); INDEX(career_role_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
roadmap_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Roadmap identifier
analysis_id	BIGINT UNSIGNED	FK	NO	analysis_runs.analysis_id	—	Analysis that produced roadmap
career_role_id	BIGINT UNSIGNED	FK	NO	career_roles.career_role_id	—	Target career role
roadmap_title	VARCHAR(255)	—	NO	—	—	Roadmap title
status	VARCHAR(30)	—	NO	—	'not_started'	not_started/in_progress/completed
created_at	DATETIME	—	NO	—	CURRENT_TIMESTAMP	Roadmap creation time

Foreign-key relationships

analysis_id → analysis_runs.analysis_id

career_role_id → career_roles.career_role_id

25. roadmap_steps

Module: 5. Learning

Purpose: Ordered learning activities within a roadmap.

Indexes / constraints: UNIQUE(roadmap_id, step_number); INDEX(skill_id); INDEX(resource_id)

Column	MySQL Type	PK / FK / UQ	NULL?	References	Default	Meaning / rule
roadmap_step_id	BIGINT UNSIGNED	PK	NO	—	AUTO_INCREMENT	Roadmap step identifier
roadmap_id	BIGINT UNSIGNED	FK	NO	learning_roadmaps.roadmap_id	—	Parent roadmap
skill_id	BIGINT UNSIGNED	FK	NO	skills.skill_id	—	Target skill
resource_id	BIGINT UNSIGNED	FK	YES	learning_resources.resource_id	NULL	Selected learning resource
step_number	INT UNSIGNED	—	NO	—	1	Order of the step
expected_days	INT UNSIGNED	YES	YES	—	NULL	Expected completion time
completion_status	VARCHAR(30)	—	NO	—	'not_started'	not_started/in_progress/completed

Foreign-key relationships

roadmap_id → learning_roadmaps.roadmap_id

skill_id → skills.skill_id

resource_id → learning_resources.resource_id

Key relationships for the ER diagram

Parent / side A	Cardinality	Child / side B	Relationship detail
users	1 : 1	candidate_profiles	candidate_profiles.user_id → users.user_id; user_id is UNIQUE in candidate_profiles.
candidate_profiles	1 : 1	candidate_preferences	candidate_preferences.profile_id → candidate_profiles.profile_id; profile_id is UNIQUE.
candidate_profiles	1 : many	education	education.profile_id → candidate_profiles.profile_id.
candidate_profiles	1 : many	experience	experience.profile_id → candidate_profiles.profile_id.
candidate_profiles	1 : many	projects	projects.profile_id → candidate_profiles.profile_id.
candidate_profiles	1 : many	certifications	certifications.profile_id → candidate_profiles.profile_id.
candidate_profiles	1 : many	resumes	resumes.profile_id → candidate_profiles.profile_id.
skills	1 : many	skill_aliases	skill_aliases.skill_id → skills.skill_id.
candidate_profiles	many : many	skills	Through candidate_skills.
career_roles	many : many	skills	Through career_role_skills.
jobs	many : many	skills	Through job_skills.
career_roles	1 : many	jobs	jobs.career_role_id → career_roles.career_role_id.
candidate_profiles	1 : many	analysis_runs	analysis_runs.profile_id → candidate_profiles.profile_id.
analysis_runs	1 : many	career_recommendations	career_recommendations.analysis_id → analysis_runs.analysis_id.
analysis_runs	1 : many	job_matches	job_matches.analysis_id → analysis_runs.analysis_id.
analysis_runs	1 : many	skill_gaps	skill_gaps.analysis_id → analysis_runs.analysis_id.
analysis_runs	1 : many	readiness_scores	readiness_scores.analysis_id → analysis_runs.analysis_id.
analysis_runs	1 : many	match_explanations	match_explanations.analysis_id → analysis_runs.analysis_id.
learning_resources	many : many	skills	Through learning_resource_skills.
analysis_runs	1 : many	learning_roadmaps	learning_roadmaps.analysis_id → analysis_runs.analysis_id.
learning_roadmaps	1 : many	roadmap_steps	roadmap_steps.roadmap_id → learning_roadmaps.roadmap_id.

Special constraint — match_explanations: the database guide says exactly one of job_match_id or career_recommendation_id should be filled for an explanation. In MySQL, implement this with a CHECK constraint if your MySQL version/enforcement setup supports it, and also validate it in the backend.

Special constraint — proficiency/required levels: the guide specifies levels from 1 to 5. The column type is TINYINT UNSIGNED; add CHECK(level BETWEEN 1 AND 5) where appropriate.

Implementation order in MySQL Workbench

1. Create the database/schema using utf8mb4.
2. Create master/reference tables first: skills and career_roles.
3. Create candidate account/profile tables.
4. Create bridge/reference tables: skill_aliases, candidate_skills, career_role_skills, jobs, job_skills.
5. Create analysis/result tables.
6. Create learning tables.
7. Insert master skills and career roles before inserting dependent records.
8. Insert the curated dummy jobs and job skills.
9. Create the EER diagram and verify every foreign-key relationship.
10. Run sample JOIN queries to confirm candidate → skills → career roles → jobs → gaps → roadmap flow.

Source alignment

The uploaded Database Design Guide defines the 25-table core database, the profile/skill/reference/analysis/learning modules, the main relationships, the dummy career/job dataset, and MySQL rules. The uploaded MCA Project document defines the application flow and expected outputs such as career recommendations, job recommendations, readiness, explainability, skill gaps, and learning roadmaps. This schema is designed to connect those requirements into one relational database.